

# İsmail SAVRAN

## Computer Engineer

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## PROFILE

Computer Engineer with a strong focus on machine learning, deep learning, and data-driven decision support systems. Experienced in developing and evaluating predictive models using large-scale real-world datasets, with particular interest in time series forecasting, natural language processing, and large language model applications. Actively involved in academic research, including IEEE-supported conference publications, and skilled in deploying end-to-end machine learning solutions using Python-based frameworks.

## EDUCATION

**Bachelor's Degree,**  
Afyon Kocatepe University

2022 – 06/2026

## SKILLS

### Programming Languages

Python, C,C#, JavaScript, SQL

### Libraries & Frameworks

Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch, TensorFlow, Flask, Gradio

### Data Science & AI

Machine Learning, Deep Learning, Data Mining, Natural Language Processing (NLP), Time Series Analysis, Predictive Modeling, Retrieval-Augmented Generation (RAG), Large Language Models (LLM)

## CERTIFICATES

- ISMSIT 2025 – IEEE Supported Conference Certificate of Participation  
Ankara, Türkiye – 14–16 November 2025 Presented a project on flight delay forecasting
- 11th Azerbaijan Certificate for Life Engineering and Applied Sciences

## PROJECTS

### Flight-Delay-Prediction-with-Machine-Learning

2025

Developed a regression-based flight delay prediction system by integrating operational flight data with meteorological features. Applied advanced data preprocessing, temporal feature engineering, and anomaly handling, followed by comparative evaluation of XGBoost, Random Forest, LightGBM, CatBoost, and LSTM models using TimeSeriesSplit and RandomizedSearchCV. Ensemble-based models demonstrated superior performance under  $R^2$ , MAE, and RMSE metrics. Model interpretability was supported through feature importance and SHAP analysis, and the final system was deployed as a Flask-based web application for real-time operational decision support.

**Deep-Learning-Based-Flight-Delay-Forecasting-with-Advanced-Time-Series-Models**

2025

Developed a multivariate time series forecasting framework for flight delay prediction by integrating operational aviation data with meteorological variables. Convolutional and attention-based deep learning approaches were employed to model nonlinear temporal dependencies, seasonality, and trends. Temporal feature engineering, including lag-based and calendar-aware representations, was applied to improve predictive performance. Models were evaluated using time series-aware validation, demonstrating robust generalization for short- and mid-term forecasting and supporting data-driven aviation operations.

**AI-Driven Phishing Detection and Categorization System**

2025

Developed a phishing detection and categorization system for email and message content using deep learning-based NLP models. The system performs binary classification (safe vs. phishing) followed by fine-grained phishing category prediction. XLM-RoBERTa was used for both tasks, augmented with rule-based URL feature extraction (e.g., IP usage, subdomains, URL shorteners, suspicious extensions). An interactive Gradio interface was implemented, with optional LLM-based (Mistral) security explanations. Models were packaged for inference without retraining, ensuring reproducibility and deployment readiness.

**Turkish Lyrical Sentiment Analysis via Fine-Tuned BERT Architecture**

Developed an end-to-end NLP pipeline for multi-class sentiment classification of Turkish song lyrics. Web-scraped data was automatically labeled using Zero-Shot Learning, followed by class imbalance handling and data engineering. A pre-trained BERT model was fine-tuned via transfer learning to capture Turkish linguistic nuances and deployed as a Flask-based RESTful API for real-time inference.

**PUBLICATIONS**

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**Forecasting Flight Delays From Operational and Weather Data: A Regression-Based Machine Learning Application** 

2025

IEEE (ISMSIT 2025)

Developed a machine learning-based flight delay prediction system by integrating large-scale U.S. domestic flight data with meteorological variables. Performed comprehensive data preprocessing, feature engineering, and time series-aware validation (TimeSeriesSplit). Evaluated multiple models including XGBoost, Random Forest, LightGBM, CatBoost, and LSTM, and deployed the best-performing model as a Flask-based web application to deliver minute-level delay predictions for operational decision support.

**FORECASTING FLIGHT DELAYS USING DEEP LEARNING-BASED TIME SERIES MODELS: A COMPARATIVE ANALYSIS OF TRANSFORMER AND CONVOLUTIONAL ARCHITECTURES**

2025

Contributed to a study on flight delay prediction using deep learning-based multivariate time series models. The research involved integrating large-scale U.S. domestic flight data with meteorological variables, followed by systematic data preprocessing and feature engineering. Time-dependent validation was conducted using TimeSeriesSplit, and representative convolutional and attention-based architectures were evaluated to analyze their ability to model temporal patterns and support operational decision-making.

**INTERESTS**

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**Volleyball Referee** — Active volleyball referee affiliated with TVF (Turkish Volleyball Federation)

**PROFESSIONAL EXPERIENCE**

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**Software Development Engineering Intern**

01/2026 – 06/2026

JANTSA JANT SANAYİ VE TİCARET A.Ş.

Designed and independently developed three production-grade systems: a corporate portal with AD/LDAP authentication and role-based access control (Node.js/Express, React, MSSQL), an IT support ticket system with email integration and Office 365 OAuth2 (React, Express), and a meeting room reservation platform.

**Intern Engineer**

06/2025 – 09/2025

JANTSA JANT SANAYİ VE TİCARET A.Ş.

Contributed to internal software development processes, gaining hands-on experience with enterprise systems, database management (MSSQL), and Active Directory integration in a manufacturing environment.

**Intern Engineer**

06/2024 – 08/2024

OKT Trailer

Assisted engineering teams with technical tasks and gained foundational exposure to industrial software and process workflows.